

Luka D. Stojiljković

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Education

BSc	School of Computing RAF , Union University in Belgrade, Computer Science	2023 – Expected 2026
BSc	Mathematical Faculty, Belgrade University, Computer Science undergraduate	2018 – 2023

Work Experience

Software Developer – AI in AEC (AEC AI consulting startup) May 2025 – January 2026

- Developed two web applications: **Neostack**, a prototype web application that helps firms catalog their software stack and license usage; **AI in AEC Flow**, an internal platform for time-task and revenue tracking, for better operational performance.

Student Projects

Software Engineering: *Java, Spring Boot, JPA/Hibernate, JWT, PostgreSQL, JUnit, Mockito, Docker, React, TypeScript, Vite, Tailwind, Vitest, Cypress, Kotlin, Jetpack Compose, Ollama, RAG*

- Led a 25-person team to deliver a full-stack enterprise banking platform across backend, web, mobile, and DevOps, and shipped a fully self-hosted autonomous LLM banking assistant on top.

Artificial Intelligence, Machine Learning, and Deep Learning: *Python, PyTorch, TensorFlow, Keras, NumPy, scikit-learn, Hugging Face Transformers, Wav2Vec2, OpenCV, MATLAB*

- Built across the modern ML stack – from-scratch neural-network and Word2Vec implementations in pure NumPy, classical ML pipelines, multimodal PyTorch fusion architectures, MATLAB CNNs with transfer learning, fine-tuned Wav2Vec2 for real-time Serbian ASR, and a fully local multi-agent ReAct LLM research system.

Misc: *Kotlin, Jetpack Compose, NFC HCE, C++17, MATLAB*

- Side work spanning mobile, systems, and graphics – an offline-first NFC business-card Android app, a complete C++17 compiler and stack-machine VM for a custom C-like language, and a from-scratch MATLAB image-processing suite.

Projects

Stem Agent – Self-improving LLM agent with evolutionary tool composition: **Python, LM Studio, Gemma, MAP-Elites, Beam Search, Typed Tool Pipelines**

- LLM agent that “differentiates” into domain experts by evolving typed tool pipelines via beam search and MAP-Elites quality-diversity archiving; promoted composites persist across sessions as inherited skills. Runs fully local on LM Studio.

PerfWindow – Real-time Windows hardware monitor: **Rust, eframe/egui, C#, .NET 8, LibreHardwareMonitor, Inno Setup**

- Two-process design: a Rust egui dashboard owns the lifecycle of a self-contained .NET 8 sensord that streams CPU, GPU, RAM, storage, fan, voltage, and network sensors via LibreHardwareMonitor; Inno Setup installer handles WinRing0 driver placement, Defender exclusion, admin elevation, and clean uninstall.

Technical Skills

Programming Languages: Java, Kotlin, Python, TypeScript, JavaScript, C, C++, Rust, SQL, MATLAB / Octave, HTML, CSS, Bash, LaTeX

Databases: PostgreSQL, MySQL, Oracle SQL, SQLite, H2, ChromaDB

Backend and Web: Spring Boot, Spring Security, Spring Data JPA, Hibernate, JWT, JUnit 5, Mockito, AssertJ, JaCoCo, OpenAPI/Swagger, PDFBox, Lombok, Maven, Gradle, FastAPI, Flask, REST APIs, Microservices, SSE

Frontend and Mobile: React 19, Vite, Tailwind CSS, shadcn/ui, Radix UI, React Router, React Hook Form, Zod, Axios, Recharts, Vitest, Cypress, Jetpack Compose, Material 3, Hilt, Retrofit, OkHttp, Moshi, Room, DataStore, Coil, Kotlin Coroutines, JavaFX, Swing

ML / AI: PyTorch, torchvision, torchaudio, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Hugging Face Transformers, Wav2Vec2, sentence-transformers, ChromaDB, LangChain, Ollama, Gemma, LM Studio, Multi-Agent LLMs, ReAct, RAG, MATLAB Deep Learning Toolbox

Tools and DevOps: Git, GitHub, Docker, docker-compose, nginx, NVIDIA Container Toolkit, CUDA, WSL, ANTLR4, Inno Setup, PowerDesigner, Google Colab, Android Studio, GitHub Pages, Microsoft Office

Additional Information

Languages:

- **Serbian:** Native
- **English:** Excellent proficiency

Achievements:

- Karate Black Belt (1st Dan Shodan)
- Diploma from Elementary Music School (Piano)
- B category driver's license